Learning Equilibrium Effort in Rank-Order Tournaments: Policy-Gradient Self-Play with Exploitability Verification

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***Abstract*—Rank-order tournaments provide a canonical model of strategic effort decision under noisy relative-performance evaluation, but equilibrium analysis can be increasingly difficult as competition becomes heterogeneous or strategically richer. This paper investigates whether policy-gradient self-play can recover analytical effort patterns and effort-policy profiles with low estimated exploitability using only simulated tournament outcomes, without access to closed-form winning probabilities, analytical payoff gradients, or equilibrium solutions. Specifically, we propose Tournament Equilibrium Learning via Proximal Policy Optimization (TEL-PPO), which trains a shared mean–concentration Beta policy through PPO self-play and evaluates unilateral deterministic effort deviations using Monte Carlo simulation, to approximate the equilibrium effort. The estimated exploitability governs termination of the training.**

We choose analytically tractable one-stage Lazear–Rosen tournaments as controlled validation settings. Using a fixed hyperparameter specification, we evaluate separately trained TEL-PPO policies across symmetric two-player, three-player, heterogeneous-cost, and heterogeneous-ability tournaments. The learned policy-implied mean efforts show close agreement with the analytical benchmarks and preserve the theoretically predicted effort ordering under cost heterogeneity. Independent post-training evaluation further shows low estimated exploitability of the full stochastic policy profiles across the evaluated configurations. Because neither learning nor verification requires the analytical solutions, this one-stage study provides a controlled validation baseline for subsequent research on learning and verification in multi-stage tournaments in which analytical equilibrium benchmarks may be unavailable.

***Index Terms*—Rank-order tournaments, equilibrium effort, multi-agent reinforcement learning, policy-gradient self-play, exploitability verification**

I. Introduction

Rank-order tournaments are a canonical model of strategic effort under uncertainty, in which players exert costly effort to improve their relative performance and rewards are assigned by rank rather than by absolute output. Since the seminal work of Lazear and Rosen [1], tournament models have provided a widely used framework for studying incentive design.

However, closed-form equilibrium characterizations typically rely on restrictive assumptions such as static interaction, simple noise structures, and symmetric players. Multi-stage competition, heterogeneous contestants, asymmetric costs of effort, and richer strategic dependencies can make analytical equilibrium derivation complicated or even untractable. These limitations motivate computational methods for approximating equilibrium behavior without requiring problem-specific closed-form solutions.

Self-play reinforcement learning is a natural candidate for this task because agents can learn directly from strategic interactions, sampled outcomes, and realized rank-based rewards. In rank-order tournaments, however, training stability alone does not establish equilibrium: a seemingly stable learned profile may still admit profitable unilateral deviations. Learning-based equilibrium computation therefore benefits from explicit unilateral-deviation assessment in addition to conventional training diagnostics. We use Proximal Policy Optimization (PPO) because it provides a stable optimization framework for stochastic continuous-action policies [2].

We propose Tournament Equilibrium Learning via Proximal Policy Optimization (TEL-PPO), a learning-and-verification framework, to approximate equilibrium-effort in rank-order tournaments. TEL-PPO represents a one-stage tournament as a bounded continuous-action stochastic game. Its key design principle is to separate policy learning, equilibrium-quality assessment, and analytical benchmarking: self-play learns the policy, unilateral-deviation evaluation governs termination, and analytical solutions are used only for post hoc validation.

This paper makes three contributions to the literature. First, we formulate equilibrium-effort approximation in noisy rank-order tournaments as a bounded continuous-action self-play problem with rewards generated from sampled rankings. Second, we introduce TEL-PPO, which couples continuous-action PPO self-play with scheduled unilateral-deviation evaluation and uses estimated exploitability to govern training termination. Third, we evaluate TEL-PPO policies across symmetric two-player, three-player, heterogeneous-cost, and heterogeneous-ability tournaments under a common experimental protocol, separately assessing analytical effort benchmark recovery and independent post-training exploitability. Overall, the learned policies recover the main analytical effort patterns across the evaluated configurations while maintaining low estimated exploitability.

II. Related Work

A. Rank-Order Tournaments and Contest Models

Contest theory studies strategic effort under competitive reward allocation. Canonical models include the Tullock rent-seeking contests [3], all-pay contests [4], and the Lazear–Rosen rank–order tournament [1]. The Lazear–Rosen model is particularly relevant to this study because rewards depend on relative performance while realized performance is affected by stochastic shocks. This structure makes effort strategically interdependent while retaining an analytically tractable equilibrium benchmark in the canonical one-stage setting [1].

Analytical tractability generally declines when contests introduce heterogeneous participants, asymmetric costs, multiple players, or richer information structures. Existing theory derives sharp results under particular structural assumptions. Franke et al. [5], for example, derive the optimal bias of a contest rule for maximizing total effort in a class of n-player contests with heterogeneous players. Bastani et al. [6] establish the existence of symmetric equal-effort equilibria for a broad class of two-player contests in which players may have heterogeneous skill distributions. These studies demonstrate the strength of analytical methods, but neither study develops a general simulation-based framework for learning equilibrium effort and evaluating resistance to unilateral deviations when an analytical characterization is unavailable.

Experimental and field research further shows that behavior in contests is sensitive to tournament design. Laboratory evidence shows that effort can vary with prize structure, tournament size, and the proportion of winners [7], [8], [9]. Field studies similarly find that competitive performance may depend on the broader contest environment, including the presence of a dominant competitor and the size of prize spreads [10], [11]. Related experimental work compares winner-take-all, lottery, and proportional-prize contests and documents distinct effort and payoff patterns across these contest formats [12]. These studies establish the behavioral importance of tournament design, whereas our focus is computational: learning and verifying approximate equilibrium-effort policies from simulated strategic interaction.

B. Reinforcement Learning for Equilibrium Computation

Multi-agent reinforcement learning (MARL) provides a flexible framework in which strategic behavior can emerge through repeated interaction, self-play, and adaptation to other agents. A broad survey of MARL methods documents approaches for cooperative and competitive environments and highlights challenges arising from nonstationarity, partial observability, and high-dimensional state and action spaces [13].

Learning-based equilibrium methods have been studied in several game classes. Reinforcement-learning approaches have been developed for two-player zero-sum simultaneous-action games [14] and for numerical recovery of Nash equilibria in continuous-action economic competition [15]. Related work uses multi-agent deep reinforcement learning to approximate equilibria in electricity-market bidding [16], combines fictitious play with deep reinforcement learning in imperfect-information games [17], and applies multi-agent learning and simulation to the design of contests and all-pay auctions [18]. Other approaches develop reinforcement-learning algorithms for multiplayer matrix games, study Stackelberg–Nash equilibria in general-sum Markov games, and examine broader connections between game theory and multi-agent learning [19]–[21].

Despite this broad literature, noisy rank-order tournaments have received comparatively limited attention as a setting for learning-based equilibrium computation. Unlike contest formulations that specify winning probabilities directly as smooth functions of effort, Lazear–Rosen tournaments generate realized rank-based rewards through stochastic relative performance. Moreover, existing learning-based equilibrium methods rarely combine convergence diagnostics with explicit unilateral-deviation tests in noisy rank-order tournaments. To the best of our knowledge, prior work has not studied policy-gradient self-play for equilibrium-effort learning in noisy rank-order tournaments with explicit unilateral-deviation verification. TEL-PPO is designed to address this gap.

III. Tournament Model and Canonical Analytical Benchmark

We begin with the canonical one-stage, two-player Lazear–Rosen rank-order tournament [1], which provides an analytical benchmark for evaluating TEL-PPO. Two symmetric, risk-neutral players simultaneously choose effort levels $e_i \in [0, e_{\max}]$. Player i's realized performance depends on both effort and luck:

$$y_i = e_i + \varepsilon_i, \qquad \varepsilon_i \overset{\text{i.i.d.}}{\sim} \mathrm{U}[-q, q], \tag{1}$$

where $q > 0$ controls the magnitude of the performance noise. Higher effort increases a player's winning probability, while the realized ranking also depends on the independently drawn performance shocks.

The player with the higher realized performance receives w_H, and the other player receives w_L, where $\Delta w \equiv w_H - w_L > 0$. With quadratic effort cost $c(e_i) = ke_i^2$, where $k > 0$, player i's expected payoff is

$$U_i(e_i, e_j) = w_L + \Delta w \, \Pr(y_i \geq y_j) - ke_i^2. \tag{2}$$

At a symmetric effort profile, the marginal effect of effort on the winning probability is

$$\left. \frac{\partial \Pr(y_i \geq y_j)}{\partial e_i} \right|_{e_i = e_j} = \frac{1}{2q}. \tag{3}$$

Therefore, for an interior solution $0 < e^* < e_{\max}$, the first-order condition yields the symmetric equilibrium effort [1]

$$e^* = \frac{\Delta w}{4qk}. \tag{4}$$

All experimental parameterizations considered below yield interior benchmark efforts satisfying $0 < e^* < e_{\max}$. The analytical benchmarks are used only for post hoc comparison;

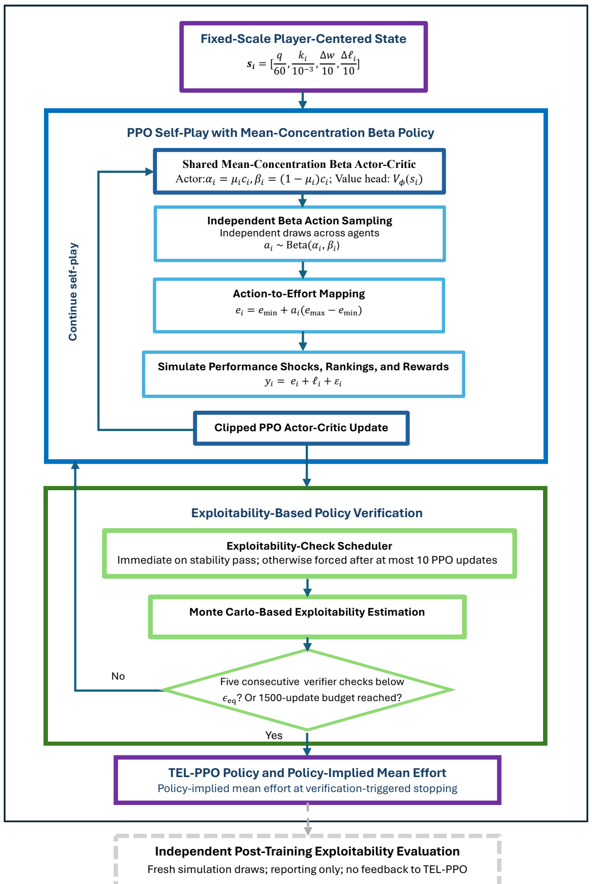

Fig. 1. TEL-PPO learning-and-verification framework. PPO self-play learns a shared mean–concentration Beta policy, while scheduled exploitability checks based on Monte Carlo deviation evaluation govern stopping. A separate evaluator reports post-training exploitability.

learning and verification rely on simulated outcomes and realized rewards. The benchmarks for the three-player and heterogeneous variants are introduced with their experimental configurations in Section V-B.

IV. Methodology

A. TEL-PPO Framework Overview

TEL-PPO combines PPO self-play using a shared mean–concentration Beta policy with scheduled exploitability checks based on Monte Carlo deviation evaluation, as summarized in Fig. 1. Given a player-centered tournament state, the shared actor–critic produces bounded effort distributions, from which players sample independently. PPO updates use sampled shocks, rankings, and realized rewards.

Rolling stability diagnostics trigger verifier calls, while a timeout forces a check after at most 10 PPO updates without one. Training stops after five consecutive checks below the prescribed tolerance. When this criterion is met, the resulting stochastic policy $\hat{\pi}^{\mathrm{TEL}}$ and its policy-implied mean effort $\hat{\mathbf{e}}^{\mathrm{TEL}}$ constitute the TEL-PPO output. A separate post-training evaluator reports final exploitability.

B. PPO Self-Play with a Mean–Concentration Beta Policy

Each tournament player is modeled as an RL agent. The agents share the policy parameters but observe player-specific states and sample their actions independently.

State representation. In the one-stage setting, player i observes the fixed-scale, player-centered state

$$\mathbf{s}_i = \left[\frac{q}{60}, \frac{k_i}{10^{-3}}, \frac{\Delta w}{10}, \frac{\Delta \ell_i}{10}\right], \qquad \Delta w = w_H - w_L, \quad (5)$$

where $\Delta \ell_i = \ell_i - \bar{\ell}_{-i}$, ℓ_i is player i's additive ability parameter, and $\bar{\ell}_{-i}$ is the mean ability of player i's opponents. Thus, $\Delta \ell_i$ measures relative ability and equals zero when players have identical abilities. This representation provides a common interface for symmetric, multi-player, heterogeneous-cost, and heterogeneous-ability environments while retaining player-specific cost and relative-ability information.

Policy representation. All players use the shared stochastic policy $\pi_\theta(a_i \mid \mathbf{s}_i)$ while sampling their actions independently. The shared policy head outputs two unconstrained scalars, $z_{\mu,i}$ and $z_{c,i}$, which parameterize the policy mean and concentration of the Beta distribution, respectively. They are transformed as

$$\mu_i = \sigma(z_{\mu,i}), \qquad c_i = 100 + \mathrm{softplus}(z_{c,i}). \quad (6)$$

The fixed offset of 100, held constant across all experiments, prevents extremely diffuse effort policies. The corresponding Beta parameters are

$$\alpha_i = \mu_i c_i, \qquad \beta_i = (1 - \mu_i) c_i. \quad (7)$$

Player i then samples

$$a_i \sim \mathrm{Beta}(\alpha_i, \beta_i), \qquad e_i = e_{\min} + a_i(e_{\max} - e_{\min}). \quad (8)$$

Players with identical states therefore have the same action distribution, although their realized efforts may differ because their actions are sampled independently. Player-specific state inputs allow the shared policy to produce different effort distributions in heterogeneous environments.

Tournament outcome and reward. Players choose their efforts simultaneously. Realized performance is

$$y_i = e_i + \ell_i + \varepsilon_i, \qquad \varepsilon_i \overset{\text{i.i.d.}}{\sim} \mathrm{U}[-q, q], \quad (9)$$

where ability enters as an additive performance offset and $\ell_i = 0$ in environments without ability heterogeneity. The realized reward is

$$r_i = \begin{cases} w_H - k_i e_i^2, & \text{if } y_i = \max_j y_j, \\ w_L - k_i e_i^2, & \text{otherwise.} \end{cases} \quad (10)$$

PPO updates are therefore based solely on sampled shocks, realized rankings, and the resulting rank-based rewards.

TEL-PPO effort estimate. TEL-PPO learns a stochastic policy, whereas the analytical benchmarks are deterministic effort profiles. For comparison, we summarize each verified stochastic policy by its policy-implied mean effort. For the verified policy parameters $\hat{\theta}$, this mean is given by

$$\hat{e}_i^{\mathrm{TEL}} = \mathbb{E}_{e_i \sim \pi_{\hat{\theta}}(\cdot \mid s_i)}[e_i] = e_{\min} + (e_{\max} - e_{\min}) \mu_{\hat{\theta}}(s_i). \quad (11)$$

The vector $\hat{\mathbf{e}}^{\mathrm{TEL}}$ is reported as a summary of the learned policy for comparison with the analytical effort benchmarks. Exploitability is evaluated for the full learned stochastic policy profile rather than for its mean-effort representation. No post-training refinement is applied.

PPO update. The shared actor–critic is optimized using the standard clipped PPO objective [2]

$$\mathcal{J}(\theta, \phi) = \mathcal{J}^{\mathrm{clip}}(\theta) - c_v \mathcal{L}^{\mathrm{value}}(\phi), \tag{12}$$

where θ and ϕ denote the actor and value-network parameters, respectively, and $c_v = 0.5$. No explicit entropy bonus is included; exploration arises from independent stochastic sampling under the Beta policy. Because each episode contains a single decision followed by a terminal reward, the advantage estimate reduces to

$$\hat{A}_i = r_i - V_\phi(\mathbf{s}_i). \tag{13}$$

The remaining optimization and network settings are reported in Table II.

C. Exploitability-Based Policy Verification

TEL-PPO evaluates the learned policy profile by estimating the maximum payoff gain available from a unilateral deviation. The verification procedure checks whether this estimated exploitability falls below the prescribed tolerance; it does not certify an exact Nash equilibrium.

Verification scheduling. Rolling KL statistics and effort drift are used to schedule verifier calls. A stability pass triggers an immediate call, while a timeout forces a call whenever 10 PPO updates have elapsed since the previous one. Thus, the stability diagnostics determine when verification occurs but are not themselves part of the exploitability criterion. Passing the stability conditions alone does not imply low exploitability. The diagnostic window and thresholds are reported in Table II.

Exploitability estimation. Let $\boldsymbol{\pi} = (\pi_1, \ldots, \pi_n)$ denote a policy profile. Player i's exploitability is defined as the maximum expected payoff gain from a unilateral deterministic effort deviation:

$$\mathrm{Exploit}_i(\boldsymbol{\pi}) = \max_{e_i' \in \mathcal{E}} \left[U_i(e_i', \boldsymbol{\pi}_{-i}) - U_i(\pi_i, \boldsymbol{\pi}_{-i}) \right], \tag{14}$$

where $\mathcal{E} = [e_{\min}, e_{\max}]$ and U_i denotes player i's expected payoff. The profile-level exploitability is

$$\mathrm{Exploit}(\boldsymbol{\pi}) = \max_i \mathrm{Exploit}_i(\boldsymbol{\pi}). \tag{15}$$

Restricting unilateral deviations to deterministic efforts is sufficient in the one-step setting because, against a fixed opponent policy, the payoff of any mixed deviation is an expectation over deterministic-effort payoffs and therefore cannot exceed the best deterministic deviation. Monte Carlo simulation is used to estimate the payoff under the current policy and each candidate deviation, while a coarse-to-fine search approximates the maximization over effort. Within each unilateral-deviation search, the same Monte Carlo sample paths are reused across candidate efforts to reduce the variance of estimated payoff differences. The resulting estimate, $\widehat{\mathrm{Exploit}}(\boldsymbol{\pi})$, is a finite-sample, grid-based estimate rather than a formal upper bound on continuous-action exploitability.

TABLE I
TOURNAMENT ENVIRONMENT CONFIGURATIONS.

Scenario	Configuration
Symmetric two-player	$n = 2$; $(w_H, w_L) = (6.5, 3.0)$; $k_i = 5.5 \times 10^{-4}$; $\ell_i = 0$; $q \in \{35, 45, 55\}$.
Three-player	$n = 3$; $(w_H, w_L) = (6.5, 3.0)$; $k_i = 1.0 \times 10^{-3}$; $\ell_i = 0$; $q \in \{35, 55\}$.
Heterogeneous-cost	$n = 2$; $(w_H, w_L) = (8.0, 5.5)$; $(k_1, k_2) = (4.0, 5.5) \times 10^{-4}$; $\ell_i = 0$; $q \in \{35, 55\}$.
Heterogeneous-ability	$n = 2$; $(w_H, w_L) = (6.5, 3.0)$; $k_i = 5.0 \times 10^{-4}$; $(\ell_1, \ell_2) = (10, 5)$; $q \in \{35, 55\}$.
Common settings	$e_i \in [e_{\min}, e_{\max}] = [0, 100]$; $\varepsilon_i \overset{\text{i.i.d.}}{\sim} \mathrm{U}[-q, q]$.

Verification-triggered stopping. At each eligible checkpoint, TEL-PPO tests whether

$$\widehat{\mathrm{Exploit}}(\boldsymbol{\pi}) \leq \epsilon_{\mathrm{eq}}. \tag{16}$$

A failed check resets the pass counter. Training terminates after five consecutive successful checks or when the budget of 1500 PPO updates is reached. We use $\epsilon_{\mathrm{eq}} = 0.01$ and $M_{\mathrm{ver}} = 65{,}536$ draws per check. A policy is called "verified" only if it satisfies the five-check criterion; this designation does not imply exact Nash equilibrium.

D. Independent Post-Training Evaluation

After stopping, the reported policy profile $\hat{\pi}^{\mathrm{TEL}}$ is evaluated using $M_{\mathrm{final}} = 200{,}000$ fresh draws and a grid over $[0, 100]$ with spacing $\Delta e = 0.25$. The reported metric is

$$\max_i \widehat{\mathrm{Exploit}}_i^{\mathrm{final}}\big(\hat{\boldsymbol{\pi}}^{\mathrm{TEL}}\big).$$

The evaluation draws are disjoint from training and training-time verification and do not influence learning, checkpoint selection, or stopping.

V. Experimental Design and Results

We evaluate TEL-PPO along three dimensions: learning dynamics and verification-triggered stopping, recovery of analytical effort benchmarks across tournament configuration, and independently post-training exploitability evaluation. All reported effort values are policy-implied means of the learned TEL-PPO policies; no post-training refinement is applied.

A. Tournament Configurations

Table I summarizes four tournament scenarios and nine environment configurations. The symmetric two-player scenario uses $q \in \{35, 45, 55\}$, while the three-player and heterogeneous scenarios use $q \in \{35, 55\}$. Effort is bounded to $[0, 100]$, which contains all analytical benchmarks considered in the experiments.

TABLE II
TEL-PPO TRAINING, VERIFICATION, AND EVALUATION SETTINGS.

Parameter	Value
Optimization	
Optimizer and learning rate	Adam; linear schedule $5 \times 10^{-5} \to 2 \times 10^{-5}$
Batch/minibatch/epochs	4096 episodes / 1024 / 1 per PPO update
Training budget and seeds	Maximum 1500 PPO updates; $N_{\text{seed}} = 5$
Policy and Network	
Clip schedule and coefficients	Linear $0.20 \to 0.15$; $c_v = 0.5$
Architecture	Shared actor–critic with two 128-unit tanh hidden layers, a mean–concentration Beta actor, and a separate value head
Beta parameterization	$\mu_i = \sigma(z_{\mu,i})$, $c_i = 100 + \text{softplus}(z_{c,i})$, $\alpha_i = \mu_i c_i$, $\beta_i = (1 - \mu_i)c_i$
State Representation	
Fixed-scale player-centered state	$\mathbf{s}_i = [q/60,\ k_i/10^{-3},\ \Delta w/10,\ \Delta\ell_i/10]$
Diagnostics and Training-Time Verification	
Stability diagnostics	Window of 20 evaluation points; rolling mean KL ≤ 0.015; rolling KL SD ≤ 0.012; effort drift ≤ 8.0
Exploitability-check schedule	Stability-triggered; otherwise forced after at most 10 PPO updates since the previous verifier call
Stopping criterion	$\widehat{\text{Exploit}}(\boldsymbol{\pi}) \leq \epsilon_{\text{eq}} = 0.01$ for five consecutive verifier checks
Verifier and deviation search	$M_{\text{ver}} = 65{,}536$ Monte Carlo samples per call; common random numbers across candidate efforts within each deviation search; coarse-to-fine grid spacings of 5.0, 1.0, and 0.25 over $[0, 100]$
Independent Post-Training Evaluation	
Evaluation protocol	$M_{\text{final}} = 200{,}000$; grid $[0, 100]$ with $\Delta e = 0.25$; fresh draws independent of training-time sampling and verification; reporting only

Note: $\Delta w = w_H - w_L$ and $\Delta\ell_i = \ell_i - \bar{\ell}_{-i}$.

Table II reports the common implementation settings. Each environment configuration is trained independently under five random seeds, with all learning, verification, and evaluation hyperparameters fixed across configurations.

B. Experimental Scenarios

We consider four one-stage tournament scenarios that test recovery of analytical effort benchmarks under different numbers of players, effort costs, and ability parameters.

Symmetric two-player tournament. The canonical two-player Lazear–Rosen tournament with identical players is the primary validation scenario. Its analytical benchmark is given by Eq. (4). We use this scenario to examine effort dynamics, verification-triggered stopping, and independent post-training exploitability across different noise levels.

Three-player tournament. This setting consists of three identical players, with one winner receiving w_H. For i.i.d. uniform shocks, the symmetric marginal winning probability remains $1/(2q)$ for any $n \geq 2$; hence Eq. (4) remains the benchmark.

Heterogeneous-cost tournament. Two players have identical ability but different effort costs, $k_1 \neq k_2$. Assuming $k_1 < k_2$, player 1 is the low-cost player, and the interior analytical benchmarks are

$$e_1^* = \frac{2k_2 q\Delta w}{8k_1k_2q^2 + (k_2 - k_1)\Delta w}, \tag{17}$$

$$e_2^* = \frac{2k_1 q\Delta w}{8k_1k_2q^2 + (k_2 - k_1)\Delta w}. \tag{18}$$

The cost coefficients are set to $(k_1, k_2) = (4.0, 5.5) \times 10^{-4}$ with $(w_H, w_L) = (8.0, 5.5)$. This setting tests whether the shared, player-conditioned policy recovers the predicted ordering in which the lower-cost player exerts greater effort.

Heterogeneous ability. Finally, we consider two-player tournaments in which players differ in ability ℓ_i but share the same effort-cost parameter k. Realized performance follows Eq. (9). Provided that $\ell_1 > \ell_2$ and $\ell_1 - \ell_2 < 2q$, the common analytical effort benchmark is

$$e^* = \frac{[2q - (\ell_1 - \ell_2)]\Delta w}{8kq^2}. \tag{19}$$

The ability parameters are set to $(\ell_1, \ell_2) = (10, 5)$. This setting tests whether TEL-PPO recovers the common ability-adjusted effort benchmark despite the players' different relative-ability states.

C. Experimental Results and Analysis

1) Learning Dynamics and Verification-Triggered Stopping: We first examine the symmetric two-player scenario with $(w_H, w_L) = (6.5, 3.0)$, $k = 5.5 \times 10^{-4}$, and $q \in \{35, 45, 55\}$.

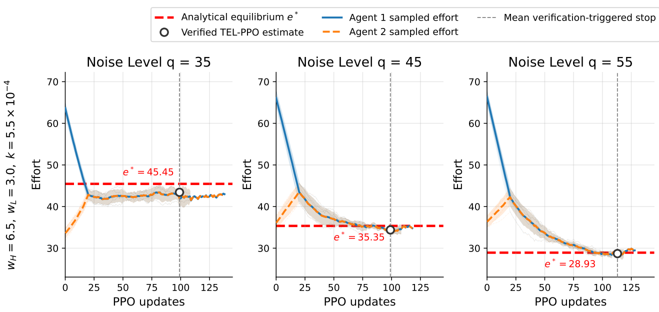

Fig. 2. TEL-PPO effort dynamics in the symmetric two-player scenario. Curves and shading show the across-seed mean sampled efforts and pointwise 95% confidence intervals. Red dashed lines mark the analytical benchmark, gray dotted lines indicate mean verification-triggered stopping updates. Open circles show the across-seed policy-implied mean efforts of the verified policies.

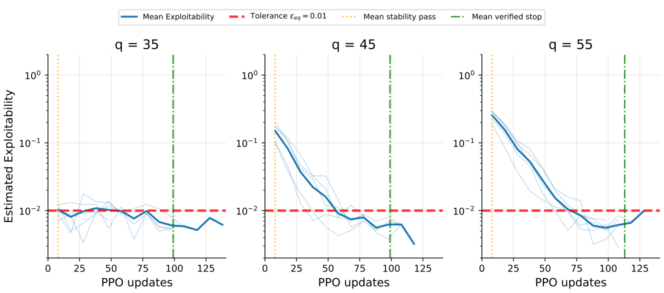

Fig. 3. Training-time exploitability during PPO self-play. Light-blue curves show seed-level estimates at verifier checkpoints, and dark-blue curves show across-seed means. No forward filling or rolling smoothing is applied. Red dashed lines mark $\epsilon_{\text{eq}} = 0.01$; orange dotted and green dash-dotted lines indicate the mean stability-pass and verification-triggered stopping updates, respectively.

Figure 2 shows the sampled-effort dynamics of the two players. Although they share the same policy parameters, independently action sampling produces distinct realized trajectories. Across all three noise levels, the trajectories move toward the analytical benchmarks.

The across-seed policy-implied mean efforts are 43.43, 34.40, and 28.70 for $q = 35$, 45, and 55, respectively, with absolute benchmark errors of 2.03, 0.95, and 0.22. The $q = 35$ configuration has largest negative deviation from the analytical benchmark.

Figure 3 reports the estimated exploitability values returned at the actual verifier checkpoints. No forward filling beyond a seed's final verifier check or rolling smoothing is applied. Each seed contributes only its observed verifier readings and terminates at its own verification-triggered stopping update. Line segments connect consecutive observations only as visual guides. The resulting sparse and jagged trajectories reflect the adaptive verification schedule and finite-sample simulation variability.

For $q = 35$, estimated exploitability is already near the prescribed tolerance at early checks but fluctuates around the threshold before the required sequence of successful checks is obtained. For $q = 45$ and $q = 55$, the checkpoint estimates exhibit an overall downward trend, although individual readings fluctuate because of finite-sample simulation noise. In all three cases, the mean stability-pass update precedes the mean verification-triggered stopping update. Thus, stability schedules verification but does not determine stopping.

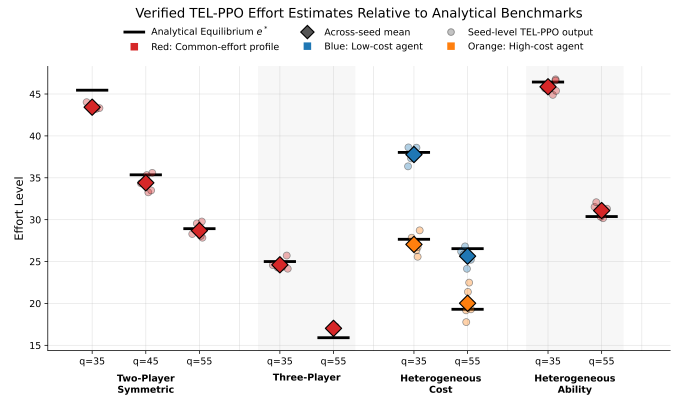

Fig. 4. Verified TEL-PPO effort estimates across tournament configurations. Small translucent circles show seed-level policy-implied effort estimates, while large filled diamonds show across-seed means. Marker color identifies the reported effort type: red denotes common-effort profiles, blue denotes low-cost agents, and orange denotes high-cost agents. Black horizontal segments indicate the corresponding analytical equilibrium efforts.

2) Independent Post-Training Evaluation: Training-time verification and post-training evaluation use disjoint simulation draws. Table III reports the independently estimated exploitability of each verified TEL-PPO policy profile. Across the nine configurations, the across-seed mean estimate ranges from 1.28×10^{-3} to 3.12×10^{-3} and remains below $\epsilon_{\text{eq}} = 0.01$ in every configuration. Thus, the low training-time estimates persist under the larger post-training evaluation.

Applying the same finite-sample evaluator to the analytical profiles produces positive evaluator baselines between 0.28×10^{-3} and 1.51×10^{-3}. These positive values arise from Monte Carlo payoff-estimation noise and maximization over a discrete deviation grid and therefore do not represent theoretical exploitability. TEL-PPO estimates are somewhat higher but remain in the low 10^{-3} range. The independent estimates provide additional evidence of low unilateral-deviation gains under the stated finite-sample deviation-evaluation protocol.

3) Effort Recovery Across Tournament Configurations: Figure 4 and Table III compare the verified TEL-PPO effort estimates with the analytical benchmarks. Across the 11 reported effort entries, the mean absolute error is 0.77, and the largest absolute error is 2.03.

In the three-player setting, the policy-implied mean efforts that TEL-PPO produced are 24.61 ± 0.64 at $q = 35$ and 17.03 ± 0.15 at $q = 55$, compared with analytical efforts of 25.00 and 15.91. The corresponding errors are 0.39 and 1.12. The $q = 55$ scenario has the largest independently estimated exploitability, $(3.12 \pm 0.72) \times 10^{-3}$, which remains below ϵ_{eq}.

In the heterogeneous-cost environments, TEL-PPO preserves the predicted effort ordering at both noise levels. At $q = 35$, the policy-implied mean efforts of the low- and high-

TABLE III
TEL-PPO RESULTS ACROSS TOURNAMENT CONFIGURATIONS.

Scenario	q	Analytical benchmark e^*	Verified TEL-PPO effort $\hat{e}_i^{\mathrm{TEL}}$	Absolute error	Independent post-training exploitability $(\times 10^{-3})$	MC evaluator baseline at e_i^* $(\times 10^{-3})$
Two-player	35	45.45	43.43 ± 0.34	2.03	2.56 ± 1.16	1.22 ± 1.20
	45	35.35	34.40 ± 1.05	0.95	1.55 ± 0.99	0.74 ± 0.45
	55	28.93	28.70 ± 0.89	0.22	1.55 ± 0.79	0.61 ± 0.38
Three-player	35	25.00	24.61 ± 0.64	0.39	2.56 ± 1.43	1.51 ± 0.46
	55	15.91	17.03 ± 0.15	1.12	3.12 ± 0.72	0.64 ± 0.39
Heterogeneous cost	35	38.03 / 27.66	37.77 ± 0.95 / 27.05 ± 1.25	0.25 / 0.61	1.51 ± 1.13	0.51 ± 0.25
	55	26.54 / 19.30	25.64 ± 1.02 / 20.02 ± 1.88	0.90 / 0.72	1.82 ± 1.33	0.28 ± 0.18
Heterogeneous ability	35	46.43	45.86 ± 0.80	0.57	2.04 ± 1.63	0.88 ± 0.54
	55	30.37	31.09 ± 0.81	0.71	1.28 ± 0.92	0.85 ± 0.41

Note: Effort and post-training exploitability estimates are reported as across-seed mean $\pm$ sample standard deviation over five training seeds. Exploitability is $\max_i \widehat{\mathrm{Exploit}}_i^{\mathrm{final}}$ under fresh simulation draws. The MC evaluator baseline applies the same finite-sample deviation-evaluation procedure to the analytical profile over five independent replications. Absolute errors are computed from unrounded analytical benchmarks and policy-implied mean efforts; displayed values are rounded to two decimal places. Heterogeneous-cost entries are ordered low-cost/high-cost. All other scenarios report the common player-level policy-implied mean effort without cross-player averaging.

cost players are 37.77 ± 0.95 and 27.05 ± 1.25, compared with analytical benchmarks of 38.03 and 27.66. At $q = 55$, the corresponding values are 25.64 ± 1.02 and 20.02 ± 1.88, compared with 26.54 and 19.30. These results show that the shared player-conditioned policy can produce asymmetric effort levels consistent with the analytical benchmarks.

In the heterogeneous-ability settings, the two players receive different relative-ability states, but their policy-implied mean efforts coincide to the reported precision. Table III therefore reports this common value rather than an average across players. The reported estimates are 45.86 ± 0.80 at $q = 35$ and 31.09 ± 0.81 at $q = 55$, yielding absolute errors of 0.57 and 0.71. These results show that the player-centered state representation can incorporate relative ability while recovering the common ability-adjusted effort benchmark.

Taken together, the effort-recovery and independent post-training exploitability results provide complementary evidence about the learned policies. Effort error quantifies the deviation of the policy-implied mean efforts from the analytical benchmarks, whereas estimated exploitability measures the payoff gains available from unilateral deviations against the full stochastic policy profile under the stated finite-sample evaluation protocol. Accordingly, proximity to e^* alone does not establish approximate best-response optimality, and low estimated exploitability does not imply exact agreement between the policy-implied mean effort and the analytical benchmark.

VI. CONCLUSION

This paper introduced TEL-PPO, a self-play framework with exploitability-based verification for learning effort policies in noisy rank-order tournaments. The framework separates policy learning from analytical benchmarking and bases training termination on empirical unilateral-deviation assessment.

Across the evaluated configurations, the learned policies reproduced the main analytical effort patterns, captured the predicted effect of cost heterogeneity, and exhibited low estimated gains from unilateral deviations under independent post-training evaluation. Together, these findings support the use of estimated exploitability as a more informative termination criterion than convergence diagnostics alone.

These findings distinguish three separate aspects of the learned outcome: stabilization of the training dynamics, recovery of the analytical effort benchmarks, and robustness of the full stochastic policy profile to unilateral deviations. Accordingly, stability diagnostics are to to schedule verification, estimated exploitability governs training termination, and analytical benchmark error is reserved for post hoc validation. This separation is intentional: the learning-and-verification procedure does not require access to an analytical equilibrium benchmark and can therefore be applied in settings where closed-form equilibrium efforts are unavailable. Because the reported values are obtained from finite Monte Carlo samples and a discrete deviation grid, they provide empirical evidence of low exploitability under the stated evaluation protocol, not a proof of exact Nash equilibrium. The analytically tractable one-stage tournaments thus serve as a controlled testbed for validating the proposed procedure. This validation represents a first step toward extending TEL-PPO to multi-stage tournaments with evolving states, sequential decisions, history-dependent policies, and higher-dimensional action spaces.

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